

- 5 (a) (magnitude of electric field strength is the potential gradient
 use of gradient at $x = 4.0 \text{ cm}$
 gradient $= 4.5 \times 10^4 \text{ NC}^{-1}$ (allow $\pm 0.3 \times 10^4$)

B1
 M1
 A1

or

$$V = \frac{Q}{4\pi\epsilon_0 x} \text{ and } E = \frac{Q}{4\pi\epsilon_0 x^2} \text{ leading to } E = \frac{V}{x} \quad (\text{B1})$$

$$\begin{aligned} E &= 1.8 \times 10^3 / 0.04 \\ &= 4.5 \times 10^4 \text{ NC}^{-1} \end{aligned} \quad \begin{array}{l} (\text{M1}) \\ (\text{A1}) \quad [3] \end{array}$$

(b) (i) $3.6 \times 10^3 \text{ V}$ A1 [1]

(ii) capacitance $= Q/V$ C1
 $= (8.0 \times 10^{-9}) / (3.6 \times 10^3)$
 $= 2.2 \times 10^{-12} \text{ F}$ A1 [2]